

Supervised Learning: Autism Spectrum Disorder (ASD) Prediction

DTSA 5509 Introduction to Machine Learning

● ● ● 2 0 2 4 F a l l 1

OUTLINE

01 Introduction

02 Data Preprocessing

03 Models

04 Results

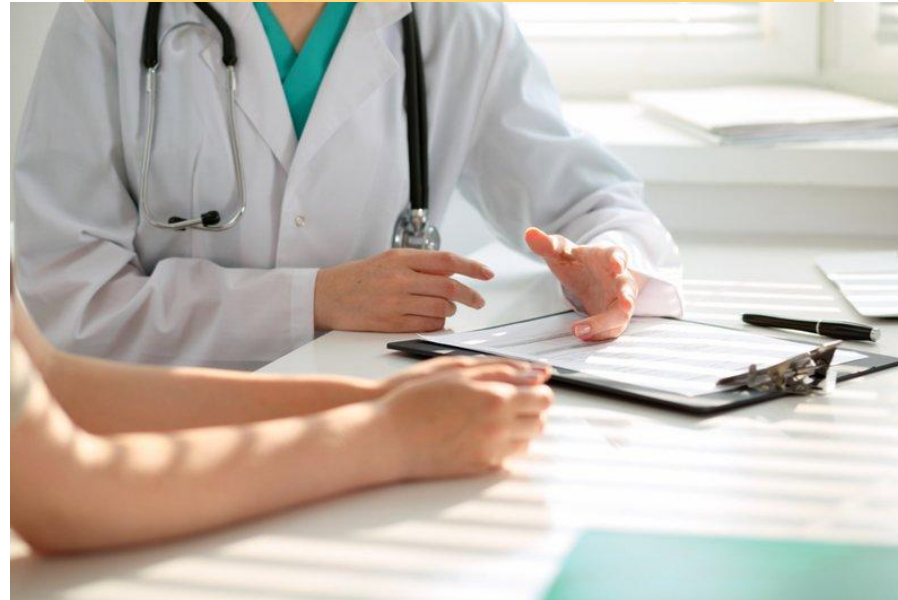
05 Discussion

06 Conclusion

01 Introduction

PROBLEM

Diagnosis of ASD is a complex and time-consuming process, requiring specialized expertise and often relying on subjective assessments. This can lead to delays in diagnosis, inconsistent results, and limited access to early intervention.



MOTIVATION

Create a cost-effective tool for early ASD diagnosis to improve individual outcomes and reduce family burdens.

APPROACH

Supervised learning algorithms are applied to predict ASD through binary classification.

02 Data Preprocessing

1985 samples 28 features

MISSING VALUES

SOCIAL_RESPONSIVENESS_SCALE

9

QCHAT_10_SCORE

39

DEPRESSION

1

SOCIAL/BEHAVIOURAL ISSUES

14

01

Consistency

Rename:

'Family member' to 'Family Member' ,
'middle eastern' to 'Middle Eastern' , ...

02

Missing Value

SimpleImputer:

Fill missing values with the **most_frequent** strategy.

03

Data Transformation

Label Encoding:

Convert string values into numerical values.

02 Data Preprocessing

04

Check for Imbalance

ASD: 1074 cases

No ASD: 911 cases

Gini Index: 0.5 (perfectly balanced)

05

Address Collinearity

Drop Columns:

Correlation matrix: correlation > 0.9

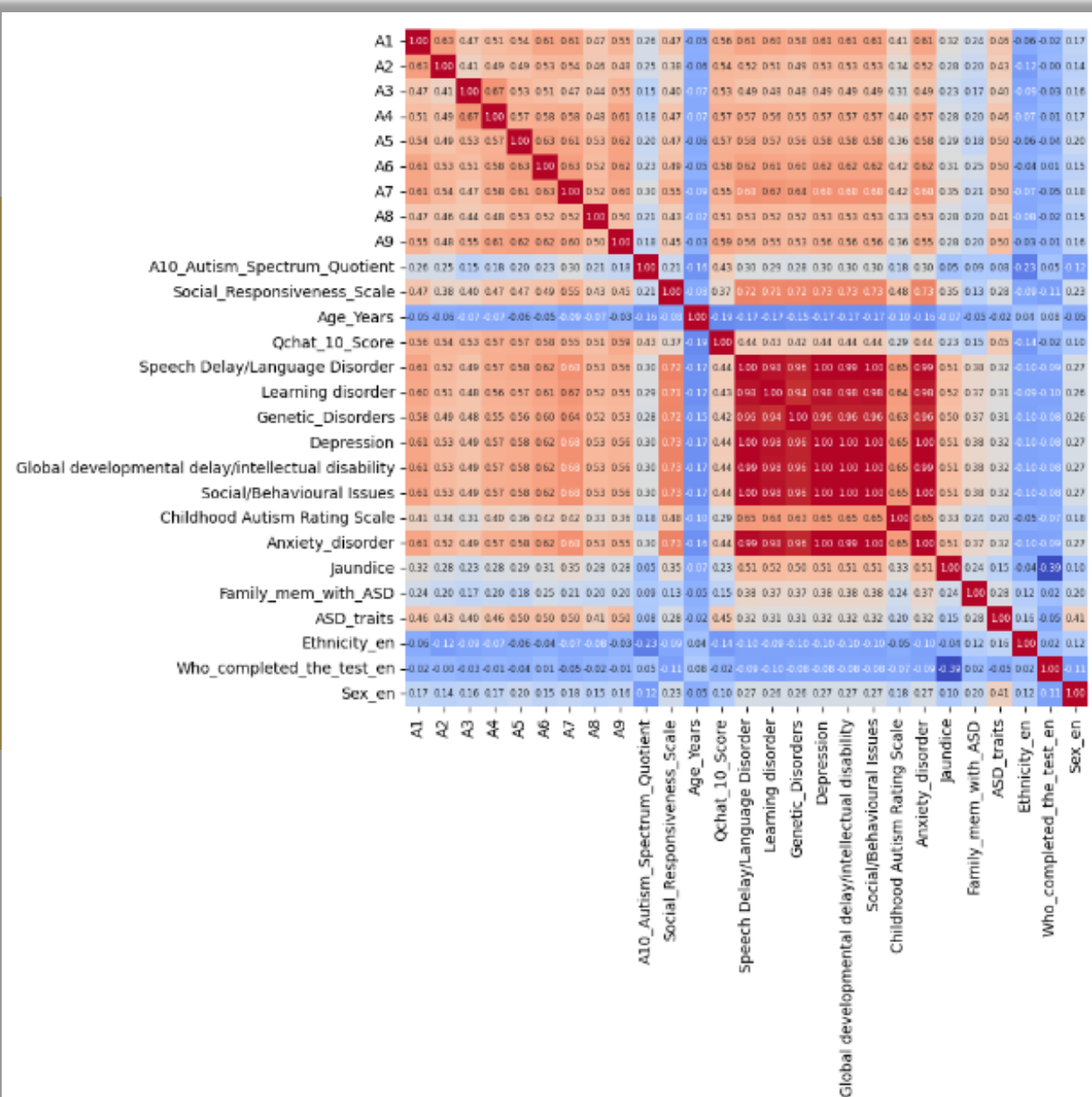
VIF values: > 10

1985

Samples

28

Features



02 Data Preprocessing

04

Check for Imbalance

ASD: 1074 cases

No ASD: 911 cases

Gini Index: 0.5 (**perfectly balanced**)

05

Address Collinearity

Drop Columns:

Correlation matrix: correlation > 0.9

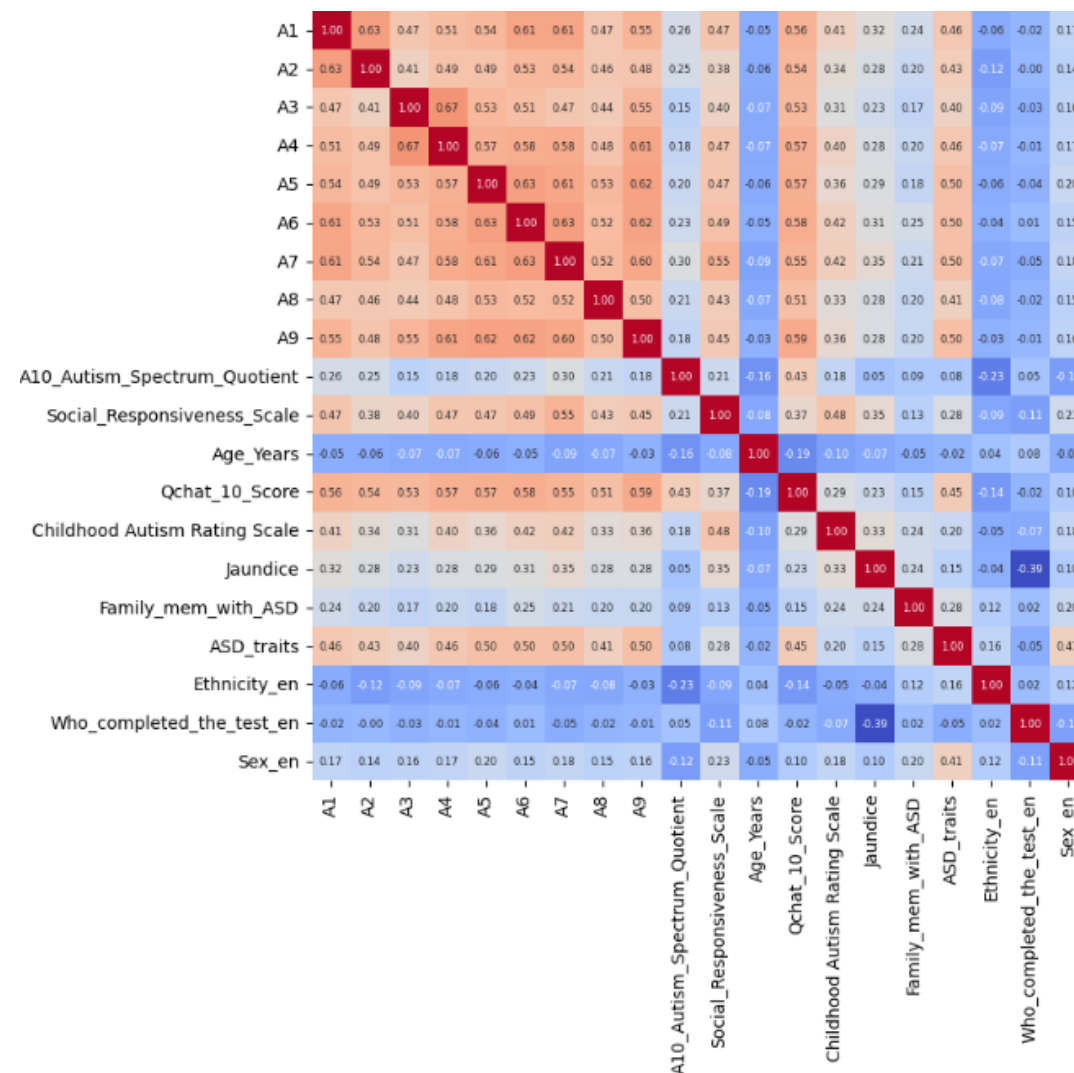
VIF values: > 10

1985

Samples

20

Features



K N N

`n_neighbors=7`
`weights='distance'`

D E C I S I O N T R E E

`max_depth=4`
`min_samples_split=2`

S V M

`C=100`
`gamma='auto'`
`kernel='rbf'`

G R A D I E N T B O O S T I N G

`n_estimators=100`

R A N D O M F O R E S T

`n_estimators=100`

A N N

16 neurons
`learning_rate=0.01`

03

Models

04 Results_1

Model	Accuracy	F1-Score	Confusion Matrix	ROC-AUC	Running Time (s)
KNN	0.96	0.96	$\begin{bmatrix} 171 & 11 \\ 6 & 209 \end{bmatrix}$	0.99	3.13
Decision Tree	0.95	0.96	$\begin{bmatrix} 171 & 11 \\ 7 & 208 \end{bmatrix}$	0.99	0.33
SVM	0.97	0.98	$\begin{bmatrix} 181 & 1 \\ 9 & 206 \end{bmatrix}$	0.98	6.46

The Performance of KNN, Decision Tree, and SVM

04 Results_2

Model	Accuracy	Confusion Matrix	Running Time (s)
Gradient Boosting	1.0	$\begin{bmatrix} 182 & 0 \\ 0 & 215 \end{bmatrix}$	0.48
Random Forest	1.0	$\begin{bmatrix} 182 & 0 \\ 0 & 215 \end{bmatrix}$	0.36
ANN	1.0	$\begin{bmatrix} 182 & 0 \\ 0 & 215 \end{bmatrix}$	17.83

The Performance of Gradient Boosting, Random Forest, and ANN

05 Discussion



Features

The feature importance analysis ranked Ethnicity highly, but it's not a direct predictor of ASD, suggesting possible overfitting and bias.



Methods

Challenges include KNN slowing with large data, Decision Tree overfitting, SVM's sensitivity to tuning, and RF, GB, and ANN being computationally expensive and harder to interpret for medical use.

Potential

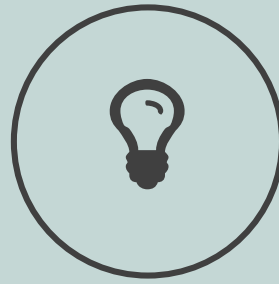
Challenges

06 Conclusion



PROJECT SUMMARY

This project predicted ASD using six machine learning methods: KNN, Decision Tree, SVM, GB, RF, and ANN.



KEY FINDINGS

For diagnosing ASD, Random Forest or Gradient Boosting strikes the best balance between high accuracy and fast runtime.



FUTURE WORK

Refine these models could involve combining behavioral scores, binning age, creating interaction terms, and using RFE or PCA for feature selection.

T H A N K S